Kichang Choi

Ph.D. Student, Civil and Environmental Engineering, Yonsei University
Founder and CEO, MANTA Robotics

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Research Interests

- Retrieval-augmented generation and multi-agent LLMs for engineering
- Embedding models for construction knowledge
- Learning-based construction robotics: imitation learning and sim-to-real RL
- Chalk-line based localization for on-site robot positioning
- Time-series forecasting for flood and levee early warning
- Vision-based construction material quality control

Education

Mar 2023 – Present **Ph.D. in Civil and Environmental Engineering**, Yonsei University, Seoul, Korea
Smart Infrastructure Lab. Advisor: Prof. Hongjo Kim. GPA 4.09 / 4.5

Mar 2013 – Feb 2019 **B.S. in Architectural Engineering**, Dankook University, Yongin, Korea
GPA 3.43 / 4.5

Experience

2026 – Present **Founder and CEO**, MANTA Robotics (Yonsei University lab startup)

- Building a mobile robot that automates interior masonry in apartment construction, using chalk-line localization and policies learned from human demonstration.
- Designed an in-house SCARA-type arm (444 mm Z axis, DM4310 joints) at about KRW 3M build cost; validated teleoperation data collection and imitation learning on a 3D-printed prototype.

Mar 2023 – Present **Graduate Researcher**, Smart Infrastructure Lab, Yonsei University

2019 – 2022 **Construction Manager**, Ssangyong E&C, Korea (3 years 6 months)

- Managed structural work schedules and trade coordination, including core-first, top-down, CFT column and TU steel beam construction.
- Led value engineering that cut about 200 t of rebar through structural redesign and about 90 t through gas pressure welding.
- Reviewed shop drawings and execution plans, ran site inspections and produced site-specific BIM models.

Publications

Journal Articles

1. Kim, S., Kim, T., **Choi, K.**, Joo, S., Kim, H. (2026). Tiny object detection using Distance-guided, Signed, and Densified Learning (DSDL) for construction site safety monitoring. *Automation in Construction*, 187, 106929. doi:10.1016/j.autcon.2026.106929
2. **Choi, K.**, Jeong, M., Kim, T., Kim, S., Baek, S., Kim, H. (2026). RAMASC: A retrieval-augmented multi-agent framework for automated structural calculation. *Advanced Engineering Informatics*, 74, 104698. doi:10.1016/j.aei.2026.104698
3. **Choi, K.**, Jeong, M., Shin, Y., Ma, J., Kim, K., Kim, H. (2026). Retrieval optimization for construction documents in low-resource languages using contrastive sentence generation and Matryoshka representation learning. *Automation in Construction*, 182, 106749. doi:10.1016/j.autcon.2025.106749
4. Idrees, S., Nugraha, J.A., Tahir, S., **Choi, K.**, Choi, J., Ryu, D.H., Kim, J.H. (2024). Automatic concrete slump prediction of concrete batching plant by deep learning. *Developments in the Built Environment*.
5. Basha, S.I., Nugraha, J.A., Rehman, A.U., **Choi, K.**, Park, S., Kim, J.H. (2024). Structuration and yield strength characterization of hybrid alkali-activated cement composites (HACC) for ultra-rapid 3D construction printing. *Construction and Building Materials*.

6. Rehman, A.U., Kang, M., Basha, S.I., **Choi, K.**, Kim, J.H. (2024). Knife cut test of concrete: The introduction of a new test method for measurement of the structural build-up of 3D concrete printing materials. *RILEM Bookseries*, Springer.
7. Rehman, A.U., Basha, S.I., **Choi, K.**, Kang, M., Kim, J.H. (2024). An analysis of penetrometer test methods for structural build-up in stiff and accelerated 3D concrete printing mixtures. Springer.

Under Review

8. **Choi, K.**, Shin, Y., Kim, Y., Osman, K., Kim, H. (2026). MANTA: Multivariate attention network for short-term water-level forecasting. Preprint, *SSRN* 6479487. Deployed as a nationwide live forecasting service

Conference Papers and Presentations

9. **Choi, K.**, Baek, S., Ma, J., Kim, H. (2025). Efficient RAG (retrieval-augmented generation) for construction in low-resource language. *Proceedings of the 42nd ISARC*, Montreal, Canada, 1575–1581. Oral. doi:10.22260/ISARC2025/0206
10. **Choi, K.**, Baek, S., Kim, H. (2024). Enhancing the performance of retrieval-augmented generation in the Korean construction domain through embedding model fine-tuning. *KICEM Annual Conference*, Jeju, Korea. Oral.
11. Chern, W.C., **Choi, K.**, Asari, V., Kim, H. (2024). *10th International Conference on Construction Engineering and Project Management* (*ICCEPM*), Sapporo, Japan. Oral.

Research Projects

2026 – Present	**Learning-based mobile masonry robot for apartment interiors** MANTA Robotics. Lead Teleoperation and first-person demonstration datasets, imitation learning, reinforcement learning in simulation (Newton, 16 parallel environments) and chalk-line based localization targeting 2 mm placement accuracy.
Aug 2024 – Present	**Autonomous evolutionary AI-based smart sensing platform for early detection and proactive response to levee failures** National Research Foundation of Korea. Project Manager Developed the MANTA water-level forecasting model; operates a nationwide real-time prediction service (sil.manta-pado.store).
2024	**Construct-RAG: contrastive sentence training and retrieval for construction documents** Project Manager Dataset generation and embedding fine-tuning that outperformed OpenAI text-embedding-3-large on Korean construction retrieval.
Mar 2023 – Present	**Eugene concrete slump prediction framework** Software and deep learning developer Optical-flow capture, CNN slump prediction and a PyQt/Docker application deployed in operating batching plants.
2023	**ABB robot-based automated vane testing for material rheology** Robot programming Camera and depth-sensor guided ABB arm performing constant-speed vane tests on construction materials.

Intellectual Property

Granted	1 Korean patent
Pending	1 Korean patent application

Skills

AI and ML	Python, PyTorch, LangChain, RAG systems, embedding fine-tuning, Axolotl, multi-agent LLMs, imitation learning, reinforcement learning
Robotics	Autodesk Fusion mechanical design, URDF, teleoperation, Newton (MJWarp) and Omniverse simulation, ABB RobotStudio
Software	Docker, PyQt, web services for real-time model deployment
Construction	Revit and BIM, construction scheduling, value engineering, site inspection